

ALGEBRAIC NUMBER THEORY

March 10, 2026

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1. DEDEKIND DOMAINS

Definition 1.0.1 (Noetherian conditions for a ring). Let R be a ring. We say:

- R is **left-Noetherian** if every ascending chain of left ideals of R stabilizes, i.e., if for any chain

$$I_1 \subseteq I_2 \subseteq I_3 \subseteq \cdots$$

of left ideals, there exists n such that $I_m = I_n$ for all $m \geq n$. That is, R satisfies the ACC on the left ideals.

- R is **right-Noetherian** if every ascending chain of right ideals of R stabilizes. That is, R satisfies the ACC on the right ideals.
- R is **Noetherian** if it is both left-Noetherian and right-Noetherian.

If R is commutative (Definition C.1.1), then R is Noetherian if and only if every ideal is finitely generated.

Definition 1.0.2 (Discrete valuation ring). A local (Definition C.1.5) integral domain $(R, \mathfrak{m})$ with maximal ideal (Definition C.1.4) $\mathfrak{m}$ is called a **discrete valuation ring (DVR)** if $\mathfrak{m}$ is principal and nonzero, and every nonzero ideal of R is of the form $\mathfrak{m}^n$ for some integer $n \geq 0$.

A **uniformizer of R** refers to any generator of $\mathfrak{m}$.

The **(normalized) discrete valuation** $v : R^\times \rightarrow \mathbb{Z}_{\geq 1}$ is given by

$$v(x) = \text{minimal } n \text{ such that } x \in \mathfrak{m}^n.$$

Alternatively, v may be extended to a map $v : R \rightarrow \mathbb{Z}_{\geq 1} \cup \{\infty\}$ by letting $v(0) = \infty$.

In fact, v extends to a discrete valuation (Definition B.2.3) on the fraction field of R by defining

$$v\left(\frac{a}{b}\right) = v(a) - v(b)$$

for $a \in R$ and $b \in R^\times$. This is a well defined map

$$v : K \rightarrow \mathbb{Z} \cup \{\infty\}.$$

Theorem 1.0.3 (Correspondence between DVRs and discretely valued fields). (**♠ TODO: define field of fractions**) Let K be a field equipped with a discrete valuation (Definition B.2.3) $v : K^\times \rightarrow \mathbb{Z}$. The valuation ring (Definition B.2.2)

$$R_v = \{x \in K \mid v(x) \geq 0\}$$

is a discrete valuation ring (Definition 1.0.2). Conversely, if $(R, \mathfrak{m})$ is a discrete valuation ring with field of fractions $K = \text{Frac}(R)$, then there exists a discrete valuation

$$v : K^\times \rightarrow \mathbb{Z}$$

such that $R = R_v$ is the valuation ring associated to v .

In particular, the category of discrete valuation rings (up to isomorphism) corresponds bijectively to the category of fields equipped with a discrete valuation.

Definition 1.0.4 (Integral element over a ring). Let R be a commutative ring with unity (Definition C.1.1).

1. Let A be an R -algebra. An element $a \in A$ is called **integral over R** if there exists a monic polynomial

$$p(x) = x^n + r_{n-1}x^{n-1} + \cdots + r_1x + r_0$$

with coefficients $r_i \in R$ such that

$$p(a) = a^n + r_{n-1}a^{n-1} + \cdots + r_1a + r_0 = 0 \quad \text{in } A.$$

2. Let A be an extension ring of R . The ring extension A/R is called an **integral extension** if every element of A is integral over R .
3. Let A be an extension ring of R . The **integral closure of R in A** , sometimes denoted by $\tilde{A}$, is the subring

$$\tilde{A} = \{a \in A : a \text{ is integral over } R\}.$$

We say that R is integrally closed in A if $\tilde{A}$ coincides with A (considered as a subring of R).

4. Let R be an integral domain with field of fractions $K = \text{Frac}(R)$. We say that R is **integrally closed** if it is integrally closed as a subring of K .

Definition 1.0.5 (Dedekind domain). An integral domain R is called a **Dedekind domain** or a **Dedekind ring** if it satisfies the following equivalent conditions: (♠ **TODO: define field of fractions**)

- R is Noetherian (Definition 1.0.1), integrally closed (Definition 1.0.4) in its field of fractions, and every nonzero prime ideal (Definition C.1.4) of R is maximal (Definition C.1.4).
- Equivalently: for every nonzero prime ideal $\mathfrak{p}$ of R , the localization (Definition C.1.3) $R_{\mathfrak{p}}$ is a discrete valuation ring (Definition 1.0.2).

The following is immediately from the definitions:

Proposition 1.0.6. Every DVR (Definition 1.0.2) is a Dedekind domain (Definition 1.0.5). Conversely, every local (Definition C.1.5) Dedekind domain is a DVR.

Proposition 1.0.7. Let R be a Dedekind domain (Definition 1.0.5). It is a PID (Definition C.1.6) if and only if it is a UFD (Definition C.1.7).

2. LOCAL AND GLOBAL FIELDS

2.1. Local fields.

Definition 2.1.1 (Local field). A **local field** is a field K with a nontrivial absolute value (Definition B.1.1) $|\cdot|$ such that K is locally compact (Definition C.2.1) under the topology induced by $|\cdot|$ (Definition C.2.4); we almost always treat a local field as a topological field with this topology.

The local field K is called **archimedean** if its absolute value is archimedean (Definition B.1.5) and is called **non-archimedean** if its absolute value is non-archimedean (Definition B.1.5).

(♠ TODO: p-adic numbers) (♠ TODO: laurent series) It is worth noting (Theorem 2.1.4)) that archimedean local fields are isomorphic to either $\mathbb{R}$ or $\mathbb{C}$ whereas non-archimedean local fields are isomorphic to finite extensions of $\mathbb{Q}_p$ or $\mathbb{F}_q((t))$ for a prime p or a prime power q .

Convention 2.1.2. Given a local field (Definition 2.1.1) K with absolute value $|\cdot|$, we almost always equip it with the metric induced by $|\cdot|$ (Definition B.1.4), which is a complete metric (Theorem 2.1.3)

Theorem 2.1.3. Let K be a local field (Definition 2.1.1). K is complete under the metric induced by (Definition B.1.4) its absolute value.

Theorem 2.1.4. (♠ TODO: TODO: define Laurent series) Up to isomorphism, local fields (Definition 2.1.1) consists precisely of:

- $\mathbb{R}$ and $\mathbb{C}$, which are the archimedean (Definition B.1.5) local fields),
- finite extensions of $\mathbb{Q}_p$ for a prime p , which are the nonarchimedean local fields of characteristic 0, and
- finite extensions of $\mathbb{F}_p((t))$, the field of formal Laurent series over a finite field, which are the non-archimedean local fields of positive characteristic.

2.2. Global fields.

Definition 2.2.1. A *number field* is a finite extension of the field of rational numbers $\mathbb{Q}$.

Definition 2.2.2. A *global function field* is a finite extension of the field of rational functions (Definition C.1.8) $\mathbb{F}_q(t)$ in one variable over a finite field $\mathbb{F}_q$.

Definition 2.2.3. A *global field* is a field K that is either:

- a number field (Definition 2.2.1), i.e. a finite extension of $\mathbb{Q}$, or
- a global function field (Definition 2.2.2), i.e. a finite extension of the field of rational functions (Definition C.1.8) $\mathbb{F}_q(t)$ in one variable over a finite field $\mathbb{F}_q$.

Definition 2.2.4. Two absolute values (Definition B.1.1) $|\cdot|_1$ and $|\cdot|_2$ on a field F are *equivalent* if there exists a positive real number $c > 0$ such that

$$|\cdot|_1 = |\cdot|_2^c.$$

Definition 2.2.5. Let K be a field. A *place of K* is some function of the form $P : K \rightarrow L \cup \{\infty\}$ where L is a field satisfying:

1. $P(0) = 0$
2. $P(1) = 1$.
3. $P(x + y) = P(x) + P(y)$
4. $P(x \cdot y) = P(x) \cdot P(y)$

where ∞ satisfies the following rules:

1. $x + \infty = \infty$ for all $x \in L$
2. $x \cdot \infty = \infty$ for $x \neq 0$.
3. $\infty + \infty$ is undefined

4. $0 \cdot \infty$ is undefined.

It should be noted that there are conventions that define $\infty + \infty$ as ∞ .

Definition 2.2.6. Let K be a field. Two places $P_1 : K \rightarrow L_1 \cup \{\infty\}$ and $P_2 : K \rightarrow L_2 \cup \{\infty\}$ are said to be equivalent if there exists a field isomorphism $\sigma : \text{im}(P_1) \cap L_1 \rightarrow \text{im}(P_2) \cap L_2$ such that $P_2 = \sigma \circ P_1$, where σ is extended to $\text{im}(P_1) \cup \{\infty\}$ by $\sigma(\infty) = \infty$.

Definition 2.2.7 (Place of a global field). Let F be a global field. A *place of F* is an equivalence (Definition 2.2.4) class (Definition 2.2.4) of absolute values (Definition B.1.1) on F .

If any (equivalently all) representatives of a place v of F is an archimedean absolute value (Definition B.1.5) (resp. non-archimedean absolute value), then we say that v is an *archimedean place* (resp. *non-archimedean place*). A representative of a place v is often denoted by $|\cdot|_v$.

Theorem 2.2.8. Let K be a global field (Definition 2.2.3). Equivalence classes (Definition 2.2.6) of places (Definition 2.2.5) of K are in bijective correspondence with equivalence classes (Definition 2.2.4) of nontrivial absolute values (Definition B.1.1) on K . In other words, defining a place (Definition 2.2.7) of a global field as an equivalence class of nontrivial absolute values is valid.

Definition 2.2.9. Let K be a global field (Definition 2.2.3) and let v be a place (Definition 2.2.7) of K . Write $|\cdot|_v$ for an absolute value (Definition B.1.1) representing v . The *completion of K at v* , often denoted K_v , is the completion of K with respect to the metric induced by $|\cdot|_v$ (Definition B.1.4).

2.3. Rings of integers of local and global fields.

Definition 2.3.1. Let R be a Dedekind domain (Definition 1.0.5) with fraction field K . Let L be a field extension of K . The *ring of integers of L over R* is the integral closure (Definition 1.0.4), denoted $\mathcal{O}_L$, of R in L .

If L/K is a finite algebraic extension, then $\mathcal{O}_L$ is itself a Dedekind domain.

Definition 2.3.2 (Ring of Integers). Let K be either a global field (Definition 2.2.3) or a nonarchimedean (Definition B.1.5) local field (Definition 2.1.1).

The *ring of integers of K* or *integer ring of K* , often denoted by notations including $\mathcal{O}_K$ and $\mathcal{O}_K$, is the set of elements of K that are integral over an appropriate base ring as follows:

- In the case where K is a number field, $\mathcal{O}_K$ is the integral closure (Definition 1.0.4) of $\mathbb{Z}$ in K , i.e.

$$\mathcal{O}_K := \{x \in K \mid x \text{ is a root of a monic polynomial } f(t) \in \mathbb{Z}[t]\}.$$

Equivalently, $\mathcal{O}_K$ is the ring of integers of K over $\mathbb{Z}$ (Definition 2.3.1).

- In the case where K is a global function field (Definition 2.2.2) over a finite field, i.e. a finite extension of $\mathbb{F}_q(t)$ (Definition C.1.8), $\mathcal{O}_K$ is the integral closure of the polynomial ring $\mathbb{F}_q[t]$ in K .

Equivalently, $\mathcal{O}_K$ is the ring of integers of K over the polynomial ring $\mathbb{F}_q[t]$ (Definition 2.3.1).

- In the case where K is a nonarchimedean (Definition B.1.5) local field (Definition 2.1.1) with discrete valuation (Definition B.2.3) v , $\mathcal{O}_K$ is the valuation ring $\mathcal{O}_v$ (Definition B.2.2) of v , i.e.

$$\mathcal{O}_K := \mathcal{O}_v = \{x \in K \mid v(x) \geq 0\}.$$

Equivalently, identifying K with (Theorem 2.1.4) a finite extension of $\mathbb{Q}_p$ or $\mathbb{F}_p((t))$, the ring $\mathcal{O}_K$ is the integral closure of $\mathbb{Z}_p$ or of $\mathbb{F}_p[[t]]$ in K . (♠ TODO:) In other words, $\mathcal{O}_K$ is the ring of integers of K over $\mathbb{Z}_p$ or over $\mathbb{F}_p[[t]]$ (Definition 2.3.1).

Theorem 2.3.3. 1. The ring of integers (Definition 2.3.2) of a nonarchimedean (Definition B.2.4) local field (Definition 2.1.1) is a DVR (Definition 1.0.2). In particular (Proposition 1.0.6), it is a Dedekind domains (Definition 1.0.5).
2. The ring of integers (Definition 2.3.2) of a global field (Definition 2.2.3) is a Dedekind domains (Definition 1.0.5).

2.4. Local fields are exactly the completions of global fields at places.

Theorem 2.4.1. Let L be a global field (Definition 2.2.3) and let v be any place (Definition 2.2.7). The completion L_v (Definition 2.2.9) of L with respect to v is a local field (Definition 2.1.1).

Remark 2.4.2. Conversely, all local fields can be obtained as completions of global fields at places.

Definition 2.4.3. Let K be a global field (Definition 2.2.3) or a local field (Definition 2.1.1), let $\mathcal{O}_K$ be its ring of integers (Definition 2.3.2), and let $I \subseteq \mathcal{O}_K$ be a nonzero ideal. The *(absolute) norm of I* is defined as the cardinality of $\mathcal{O}_K/I$, which is always finite.

Definition 2.4.4. Let K be a number field with ring of integers $\mathcal{O}_K$. Let $\mathfrak{p}$ be a nonzero prime ideal of $\mathcal{O}_K$.

1. The *(normalized) $\mathfrak{p}$ -adic valuation on K* refers to the discrete valuation (Definition B.2.3) $v_{\mathfrak{p}}$ defined by

$$v_{\mathfrak{p}}(a) = \max\{n : a \in \mathfrak{p}^n\}$$

for $a \in \mathcal{O}_K \setminus \{0\}$ and

$$v_{\mathfrak{p}}(x) = v_{\mathfrak{p}}(a) - v_{\mathfrak{p}}(b)$$

if $x = \frac{a}{b}$ for $a, b \in \mathcal{O}_K \setminus \{0\}$. This is well defined.

2. The *$\mathfrak{p}$ -adic absolute value on K* refers to the non-archimedean (Definition B.1.5) absolute value (Definition B.1.1) given by

$$|x|_{\mathfrak{p}} = N(\mathfrak{p})^{-v_{\mathfrak{p}}(x)}$$

(Definition 2.4.3).

Theorem 2.4.5. 1. Let K be a number field (Definition 2.2.1) with ring of integers (Definition 2.3.2) $\mathcal{O}_K$.

- (a) The nonarchimedean (Definition B.1.5) absolute values (Definition B.1.1) on K are all equivalent to (Definition 2.2.4) $\mathfrak{p}$ -adic absolute values (Definition 2.4.4) for nonzero prime ideals (Definition C.1.4) $\mathfrak{p}$ of $\mathcal{O}_K$.

- (b) The archimedean absolute values on K are equivalent to absolute values induced by an embedding $\sigma : K \hookrightarrow \mathbb{C}$ (among these may be embeddings whose images lie in $\mathbb{R}$).
- 2. Let K be a global function field (Definition 2.2.2) with ring of integers (Definition 2.3.2) $\mathcal{O}_K$.
All absolute values (Definition B.1.1) on K are non-archimedean (Definition B.1.5).

2.5. Ramification and inertia of extensions of local and global fields.

Definition 2.5.1. Let L/K be an extension of valued fields (Definition B.2.1) with valuations $v : K \rightarrow \Gamma_K \cup \{\infty\}$ and $w : L \rightarrow \Gamma_L \cup \{\infty\}$. We say that w is an *extension of v* if

- 1. Γ_K is an ordered subgroup of Γ_L
- 2. For all $x \in K$, $w(x) = v(x)$.

In other words, $v = w|_K$.

It is appropriate to refer to $(L, w)/(K, v)$ as an *extension of valuation fields*.

Context 2.5.2. Let L/K be an extension of fields, let $w : L \rightarrow \Gamma_L \cup \{\infty\}$, $v : K \rightarrow \Gamma_K \cup \{\infty\}$ be valuations (Definition B.2.1) on L and K with w extending (Definition 2.5.1) v , let $\mathcal{O}_w$ and $\mathcal{O}_v$ be their respective valuation rings (Definition B.2.2), which are local (Definition C.1.5), with maximal ideals (Definition C.1.4) $\mathfrak{m}_w$ and $\mathfrak{m}_v$, and let k_w and k_v be their residue fields.

Definition 2.5.3. Assume Context 2.5.2. The *ramification index of w over v* , denoted $e(w/v)$, is the index of the value group Γ_v in Γ_w :

$$e(w/v) = (\Gamma_w : \Gamma_v)$$

Definition 2.5.4. Assume Context 2.5.2. The *inertial degree index of w over v* , denoted $f(w/v)$, is the degree of the residue field extension:

$$f(w/v) = [k_w : k_v]$$

Definition 2.5.5. Assume Context 2.5.2.

- 1. Further assume that L/K is finite. The extension $(L, w)/(K, v)$ of valuation fields is *unramified* if
 - (a) $e(w/v) = 1$ (Definition 2.5.3)
 - (b) $f(w/v) = [L : K]$ (Definition 2.5.4)
 - (c) The residue field extension k_w/k_v is separable.
- 2. Removing the assumption that L/K is finite, the extension $(L, w)/(K, v)$ of valuation fields is *unramified* if for every finite subextension $K \subseteq E \subseteq L$, $(E, w|_E)/(K, v)$ is unramified in the above sense.

Definition 2.5.6. Assume Context 2.5.2.

- 1. Further assume that L/K is finite. The extension $(L, w)/(K, v)$ of valuation fields is *totally ramified* if
 - (a) $e(w/v) = [L : K]$ (Definition 2.5.3)

- (b) $f(w/v) = 1$ (Definition 2.5.4)
- 2. Removing the assumption that L/K is finite, the extension $(L, w)/(K, v)$ of valuation fields is *totally ramified* if for every finite subextension $K \subseteq E \subseteq L$, $(E, w|_E)/(K, v)$ is totally ramified in the above sense.

Definition 2.5.7. Assume Context 2.5.2 and let p be the characteristic of the residue field k_v .

- 1. Further assume that L/K is finite.
 - (a) The extension $(L, w)/(K, v)$ is *tamely ramified* if:
 - (i) The residue field extension k_w/k_v is separable;
 - (ii) If $p > 0$, then $p \nmid e(w/v)$;
 - (iii) The extension is defectless, i.e., $[L : K] = e(w/v)f(w/v)$.
 - (b) The extension $(L, w)/(K, v)$ is *wildly ramified* if it is not tamely ramified.
 - *Note:* For non-archimedean (Definition B.2.4) local fields (Definition 2.1.1), this occurs if and only if $p \mid e(w/v)$.
- 2. Removing the assumption that L/K is finite, the extension $(L, w)/(K, v)$ is *tamely ramified* (resp. *wildly ramified*) if for every (resp. for some) finite subextension $K \subseteq E \subseteq L$, $(E, w|_E)/(K, v)$ is tamely ramified (resp. wildly ramified).

3. ADÈLES AND IDÈLES OF GLOBAL FIELDS

Definition 3.0.1. Let K be a global field (Definition 2.2.3). Write M_K for the set of all places (Definition 2.2.7) of K and write M_K^∞ for the set of archimedean places (Definition 2.2.7) of K . Let $S \subseteq M_K$ be some subset of places of K (typically, S is a finite set). For each $v \in M_K$, write $\mathcal{O}_v$ for the ring of integers (Definition 2.3.2) in the completion K_v (Definition 2.2.9) (Theorem 2.4.1)

- The *adèle ring of K* , denoted $\mathbb{A}_K$, is the restricted direct product of the K_v (over all places v of K), with respect to the $\mathcal{O}_v$ at non-archimedean (Definition B.1.5) v :

$$\mathbb{A}_K = \left\{ (x_v)_v \in \prod_{v \in M_K} K_v \mid x_v \in \mathcal{O}_v \text{ for all but finitely many non-archimedean } v \right\}.$$

- The *idèle group of K* , commonly denoted $\mathbb{A}_K^\times$ or $\mathbb{I}_K$, is the group of invertible elements of $\mathbb{A}_K$:

$$\mathbb{I}_K = \mathbb{A}_K^\times = \left\{ (x_v)_v \in \prod_{v \in M_K} K_v^\times \mid x_v \in \mathcal{O}_v^\times \text{ for all but finitely many non-archimedean } v \right\},$$

where $\mathcal{O}_v^\times$ denotes the group of units of $\mathcal{O}_v$ for non-archimedean v .

- The *adèle ring outside S of K* , commonly denoted $\mathbb{A}_K^S$ or $\mathbb{A}_{K,S}$, is the restricted product of the completions K_v over all places $v \in M_K \setminus S$, with respect to the rings of integers

$\mathcal{O}_v$ at non-archimedean places:

$$\mathbb{A}_{K,S} = \mathbb{A}_K^S = \left\{ (x_v)_v \in \prod_{v \in M_K \setminus S} K_v \mid x_v \in \mathcal{O}_v \text{ for all but finitely many non-archimedean } v \right\}.$$

- The *idèle group outside S of K* , commonly denoted $(\mathbb{A}_K^\times)^S$, $(\mathbb{A}_{K,S}^\times)$, $\mathbb{I}_K^S$, or $\mathbb{I}_{K,S}$ is the group of invertible elements of $\mathbb{A}_K^S$:

$$(\mathbb{A}_K^\times)^S = \left\{ (x_v)_v \in \prod_{v \in M_K \setminus S} K_v^\times \mid x_v \in \mathcal{O}_v^\times \text{ for all but finitely many non-archimedean } v \right\}.$$

- The *ring of finite adèles of K* , commonly denoted $\mathbb{A}_{K,\text{fin}}$, $\mathbb{A}_K^{\text{fin}}$, $\mathbb{A}_{K,\text{f}}$, $\mathbb{A}_K^{\text{f}}$, is the adèle ring outside $S = M_K^\infty$, the set of archimedean places of K :

$$\mathbb{A}_{K,\text{fin}} := \mathbb{A}_K^{M_K^\infty} = \left\{ (x_v)_v \in \prod_{v \notin M_K^\infty} K_v \mid x_v \in \mathcal{O}_v \text{ for all but finitely many non-archimedean } v \right\}.$$

- The *finite idèle group of K* , commonly denoted $\mathbb{A}_{K,\text{fin}}^\times$, $\mathbb{I}_{K,\text{fin}}$, $\mathbb{I}_K^{\text{fin}}$, $\mathbb{I}_{K,\text{f}}$, $\mathbb{I}_K^{\text{f}}$ etc. is the group of units of the ring of finite adèles:

$$\mathbb{A}_{K,\text{fin}}^\times := (\mathbb{A}_K^\times)^{M_K^\infty} = \left\{ (x_v)_v \in \prod_{v \notin M_K^\infty} K_v^\times \mid x_v \in \mathcal{O}_v^\times \text{ for all but finitely many non-archimedean } v \right\}.$$

All of these are equipped with the restricted product topology induced by the topologies of the local fields (Definition 2.1.1) K_v and the subspace topologies thereof.

Definition 3.0.2 (Idelic norm of a global field). Let F be a global field. The *idelic norm* (also called the *module*) is the map

$$|\cdot| : \mathbb{A}_F^\times \rightarrow \mathbb{R}_{>0}$$

defined by (♠ TODO: definition of canonical absolute value of places)

$$|(x_p)_p| := \prod_p |x_p|_p,$$

where the product is well-defined as all but finitely many factors equal 1.

(♠ TODO: state the product formula for global fields, define idele class group) By the product formula for global fields, the idelic norm factors through the idele class group and satisfies

$$|a| = 1 \quad \text{for all } a \in F^\times,$$

where $F^\times$ is diagonally embedded in $\mathbb{A}_F^\times$.

4. CHEBOTAREV DENSITY THEOREM

The Chebotarev density theorem is a statement that Frobenius elements $\text{Frob}_{\mathfrak{p}}$ are equidistributed across the conjugacy classes of $\text{Gal}(L/K)$ of a finite Galois extension of number fields, showing that the splitting behavior of primes in extensions is governed by uniform distribution with respect to the Galois group structure.

Theorem 4.0.1 (Chebotarev Density Theorem for number fields). (♠ TODO: define unramified prime) Let L/K be a finite Galois extension (Definition A.0.1) of number fields (Definition 2.2.1) with Galois group $G = \text{Gal}(L/K)$ (Definition A.0.1). For a conjugacy class $C \subseteq G$, let

$$\pi_C(x) = \#\{\mathfrak{p} \subseteq \mathcal{O}_K : N\mathfrak{p} \leq x, \mathfrak{p} \text{ unramified in } L, \text{Frob}_{\mathfrak{p}} \in C\}$$

be the number of prime ideals $\mathfrak{p}$ of K with norm at most x whose Frobenius conjugacy class $\text{Frob}_{\mathfrak{p}}$ in G equals C .

(♠ TODO: define natural density of primes) Then the natural density of such primes exists and satisfies

$$\lim_{x \rightarrow \infty} \frac{\pi_C(x)}{\pi(x)} = \frac{|C|}{|G|},$$

where $\pi(x)$ is the number of prime ideals $\mathfrak{p} \subseteq \mathcal{O}_K$ with $N\mathfrak{p} \leq x$.

Theorem 4.0.2 (Chebotarev Density theorem for function fields, see e.g. [Ros02, Theorem 9.13A, 9.13B]). Let L/K be a Galois extension (Definition A.0.1) of global function fields (Definition 2.2.2) with Galois group $G = \text{Gal}(L/K)$. Let $C \subset G$ be a conjugacy class in G and S'_K be the set of primes of K which are unramified in L . (♠ TODO: continue statement)

APPENDIX A. GALOIS THEORY

Definition A.0.1 (Galois Extension). An extension L/K is called a *Galois extension* if it is both a normal extension and a separable algebraic extension. Its *Galois group*, usually denoted by $\text{Gal}(L/K)$, is defined to be the automorphism group $\text{Aut}(L/K)$.

APPENDIX B. ABSOLUTE VALUES AND VALUATIONS ON FIELDS

B.1. Absolute values on fields.

Definition B.1.1 (Absolute Value on a Field). Let F be a field. An *absolute value on F* is a function

$$|\cdot| : F \rightarrow \mathbb{R}_{\geq 0}$$

that satisfies the following properties for all $a, b \in F$:

1. Non-negativity: $|a| \geq 0$,
2. Positive-definiteness: $|a| = 0 \iff a = 0$,
3. Multiplicativity: $|ab| = |a| \cdot |b|$,
4. Triangle inequality: $|a + b| \leq |a| + |b|$.

Here, 0 denotes the additive identity of the field F , and the codomain $\mathbb{R}_{\geq 0}$ consists of non-negative real numbers.

Definition B.1.2 (Discrete absolute value). Let K be a field equipped with an absolute value (Definition B.1.1) $|\cdot|$. The absolute value $|\cdot|$ is called *discrete* if its image $|K^\times| = \{|x| : x \in K^\times\} \subseteq \mathbb{R}_{>0}$ is a discrete subgroup of the multiplicative group $\mathbb{R}_{>0}$ (with the usual topology).

Definition B.1.3 (Trivial absolute value). Let K be a field. The *trivial absolute value on K* is the absolute value (Definition B.1.1)

$$|\cdot|_{\text{triv}} : K \rightarrow \mathbb{R}_{\geq 0}$$

defined by

$$|x|_{\text{triv}} := \begin{cases} 0 & \text{if } x = 0, \\ 1 & \text{if } x \neq 0. \end{cases}$$

Any absolute value on K that is not equivalent (Definition 2.2.4) to this absolute value is called *nontrivial*.

Definition B.1.4. Let K be a field with an absolute value $|\cdot|$. By the metric induced by $|\cdot|$ we mean the metric induced by $|\cdot|$ (Definition C.2.2) regarded as a norm (Definition C.2.5) on the 1-dimensional K -vector space K with absolute value $|\cdot|_v$.

Definition B.1.5. Let F be a field. An absolute value on F (Definition B.1.1) is said to be *non-archimedean* if

$$|x + y| \leq \max(|x|, |y|) \quad \text{for all } x, y \in F$$

and is said to be *archimedean* otherwise.

Definition B.1.6 (Topology on a field induced by an absolute value). Let K be a field equipped with an absolute value (Definition B.1.1) $|\cdot| : K \rightarrow \mathbb{R}_{\geq 0}$. The *topology induced by $|\cdot|$ on K* is defined by declaring a subset $U \subseteq K$ to be open if for every $x \in U$, there exists $\varepsilon > 0$ such that

$$B(x, \varepsilon) := \{y \in K : |y - x| < \varepsilon\} \subseteq U.$$

The collection of all such open sets forms a topology on K , making $(K, \mathcal{T}_{|\cdot|})$ a topological field. The set $B(x, \varepsilon)$ is called the *open ball of radius ε around x* .

Equivalently, the topology induced by $|\cdot|$ on K is the topology induced by (Definition C.2.3) the metric induced by $|\cdot|$ (Definition C.2.2) as a norm (Definition C.2.5) on the 1-dimensional K -vector space K .

B.2. Valuations on fields.

Definition B.2.1 (Valuation on a field). Let K be a field. A *valuation on K* is a function

$$v : K \rightarrow \Gamma \cup \{\infty\},$$

(♠ TODO: ordered group) where $(\Gamma, +, \leq)$ is a totally ordered abelian group and ∞ is an element greater than all elements of Γ , satisfying for all $x, y \in K$:

1. $v(x) = \infty$ if and only if $x = 0$,

2. $v(xy) = v(x) + v(y)$,
3. $v(x + y) \geq \min\{v(x), v(y)\}$.

Alternatively (and essentially equivalently), a valuation on K is also defined as a function $v : K^\times \rightarrow \Gamma$ with properties (2) and (3) and is extended into a function $v : K \rightarrow \Gamma \cup \{\infty\}$ by setting $v(0) = \infty$.

The pair (K, v) is called a **valued field**.

Definition B.2.2 (Valuation Ring of a Valued Field). Let K be a field equipped with a valuation (Definition B.2.1) (**♠ TODO: totally ordered abelian group**) $v : K \rightarrow \Gamma$, where Γ is a totally ordered abelian group.

The **valuation ring of the valued field** (K, v) is the subring

$$\mathcal{O}_v := \{x \in K \mid v(x) \geq 0\},$$

where 0 is the neutral element of Γ . This ring $\mathcal{O}_v$ is a local ring (Definition C.1.5) with maximal ideal (Definition C.1.4)

$$\mathfrak{m}_v := \{x \in K \mid v(x) > 0\}.$$

Its **residue field** is defined as $\kappa_v := \mathcal{O}_v / \mathfrak{m}_v$.

Definition B.2.3 (Discrete valuation). Let (K, v) be a valued field (Definition B.2.1). The valuation v is called **discrete** if the value group $\Gamma = v(K^\times)$ is isomorphic, as an ordered group, to $\mathbb{Z}$ (the integers with the usual order).

Definition B.2.4. Let K be a field and let $v : K \rightarrow \mathbb{Z}$ be a discrete valuation (Definition B.2.3). If a positive real number $c < 1$. The function

$$|x|_v := \begin{cases} c^{v(x)} & \text{if } x \in K^\times \\ 0 & \text{if } x = 0 \end{cases}$$

is a nonarchimedean absolute value on K (Definition B.1.5). Different values of c yield equivalent absolute values (Definition 2.2.4) and thus do not change the topology induced by $|\cdot|_v$ (Definition B.1.6)

APPENDIX C. MISCELLANEOUS DEFINITIONS

C.1. Abstract algebra.

Definition C.1.1. A **commutative (unital) ring** is a ring $(R, +, \cdot)$ such that $\cdot$ is a commutative operation, i.e. $a \cdot b = b \cdot a$.

For many writers (e.g. “commutative” algebraists or number theorists), a **ring** refers to a commutative ring as above.

Definition C.1.2. Let $(R, +, \cdot)$ be a not-necessarily commutative ring.

1. An element $a \in R$ is a **left zero-divisor** if there exists a nonzero $x \in R$ such that $ax = 0$. Otherwise, a is called **left regular** or **left cancellable**.

2. An element $a \in R$ is a **right zero-divisor** if there exists a nonzero $x \in R$ such that $xa = 0$. Otherwise, a is called **right regular** or **right cancellable**.
3. An element $a \in R$ is a **zero-divisor** if it is a left zero-divisor or a right zero-divisor.
4. An element $a \in R$ is a **two-sided zero-divisor** if it is both a left zero-divisor and a right zero-divisor.
5. An element $a \in R$ is **regular**, **cancellable**, or a **non-zero-divisor** if it is both left and right regular.

A zero-divisor of any kind that is not itself 0 is said to be a **nonzero zero divisor** or a **nontrivial zero divisor** of its kind.

Definition C.1.3. Let R be a ring and let $S \subseteq Z(R)$ be a multiplicative subset contained in the center of R . The **localization of R at S** , denoted by $S^{-1}R$, is the ring whose elements are equivalence classes of pairs $(r, s) \in R \times S$ under the equivalence relation

$$(r, s) \sim (r', s') \iff \exists u \in S \text{ such that } u(sr' - s'r) = 0.$$

Write $\frac{r}{s}$ for the equivalence class of (r, s) . Addition and multiplication on representatives are defined by

$$\begin{aligned} \frac{r}{s} + \frac{r'}{s'} &= \frac{rs' + r's}{ss'}, \\ \frac{r}{s} \cdot \frac{r'}{s'} &= \frac{rr'}{ss'}. \end{aligned}$$

The map $r \mapsto \frac{r}{1}$ defines a ring homomorphism; therefore, $S^{-1}R$ is naturally an R -algebra.

Given an element $f \in Z(R)$, the localization of R at $S = \{f^n : n \geq 0\}$ is denoted by R_f .

In the case that R is a commutative ring and if P is a prime ideal of R (Definition C.1.4), then $R_P := S^{-1}R$ with $S = R \setminus P$ is called the **localization of R at P** . It is a local ring (Definition C.1.5) whose maximal ideal (Definition C.1.4) is given by

$$S^{-1}P = \left\{ \frac{p}{s} \in R_P : p \in P \right\}.$$

Definition C.1.4. Let R be a (not necessarily commutative) ring. A proper two-sided ideal $P \trianglelefteq R$ is called a **prime ideal** if the following equivalent conditions holds:

1. If I, J are left ideals and $IJ \subset P$, then $I \subset P$ or $J \subset P$.
2. If I, J are right ideals and $IJ \subset P$, then $I \subset P$ or $J \subset P$.
3. If I, J are two-sided ideals and $IJ \subset P$, then $I \subset P$ or $J \subset P$.
4. If $x, y \in R$ with $xRy \subset P$, then $x \in P$ or $y \in P$.

A proper left/right/two-sided ideal $M \subsetneq R$ is called **maximal** if there exists no other left/right/two-sided ideal $J \trianglelefteq R$ such that $M \subsetneq J \subsetneq R$. Equivalently,

- a left/right ideal M of R is maximal if and only if the quotient module R/M is a simple left/right R -module.
- a two-sided ideal M of R is maximal if and only if the quotient ring R/M is a simple ring.

Definition C.1.5. Let R be a ring with unity, not necessarily commutative. The ring R is called a *local ring* if it has a unique maximal left ideal (Definition C.1.4). In this case, R also has a unique maximal right ideal, and these coincide with the Jacobson radical $J(R)$ of R . The unique maximal left (and right) ideal of a local ring R may sometimes be denoted by $\mathfrak{m}_R$.

Definition C.1.6 (Principal ideal ring/domain (PID)). Let R be a commutative unital ring (Definition C.1.1). Then R is a *principal ideal ring (PIR)* if every ideal of R is principal. If R is additionally an integral domain, then R is said to be a *principal ideal domain (PID)*.

Definition C.1.7 (Unique factorization domain (UFD) / Factorial ring). (♠ TODO: define irreducible elements) An R is called a *unique factorization domain (UFD)* or *factorial ring* if every nonzero nonunit element of R can be factored as a product of irreducible elements uniquely up to order and units.

Definition C.1.8. Let R be an integral domain. Let $n \geq 1$ be an integer. The *function field over R in n variables $x_1, \dots, x_n$* is the field $R(x_1, \dots, x_n)$ defined as the field of fractions of the polynomial ring $R[x_1, \dots, x_n]$, which is an integral domain.

$R(x_1, \dots, x_n)$ is also called the *field of rational numbers over R in n variables*. An element of $R(x_1, \dots, x_n)$ is called a *rational function over R in n variables*.

C.2. Absolute values and norms.

Definition C.2.1 (Locally compact). Let $(X, \mathcal{T})$ be a topological space. X is *locally compact* if for every $x \in X$, there exists an open set $U \in \mathcal{T}$ containing x and a compact set $K \subseteq X$ such that $x \in U \subseteq K$.

Definition C.2.2 (Extended Metric Induced by an Extended Norm). Let V be a vector space over a field F equipped with an absolute value (Definition B.1.1)

$$|\cdot| : F \rightarrow [0, \infty),$$

and let $\|\cdot\| : V \rightarrow [0, \infty]$ be an extended norm on V (Definition C.2.5). Then the *extended metric induced by the extended norm* is the function

$$d : V \times V \rightarrow [0, \infty]$$

defined by

$$d(x, y) := \|x - y\|.$$

It is indeed an extended metric. If $\|\cdot\|$ is a norm, then d is a metric.

Definition C.2.3 (Topology induced by an extended metric). Let (X, d) be an extended metric space.

1. Given $\varepsilon \in (0, \infty)$ and $x \in X$, the set

$$B(x, \varepsilon) := \{y \in X : d(x, y) < \varepsilon\}$$

is called the *open ball of radius ε centered at x* .

2. The *topology induced by d on X* is the topology defined by declaring a subset $U \subseteq X$ to be open if for every $x \in U$, there exists $\varepsilon \in (0, \infty)$ such that $B(x, \varepsilon)$ is contained in U .

Definition C.2.4 (Topology induced by a norm). Let V be a vector space over a field K with absolute value (Definition B.1.1) $|\cdot|$, and let $\|\cdot\|$ be an extended norm on V (Definition C.2.5). The *topology induced by the extended norm $\|\cdot\|$ on V* is defined by declaring a subset $U \subseteq V$ to be open if for every $x \in U$, there exists $\varepsilon > 0$ such that

$$B(x, \varepsilon) := \{y \in V : \|y - x\| < \varepsilon\}$$

is contained in U . The set $B(x, \varepsilon)$ is called the *open ball of radius ε around x* . The collection of all such open sets forms a topology on V .

Equivalently, the topology on V induced by the extended norm $\|\cdot\|$ is the topology on V induced by (Definition C.2.3) the extended metric $d : V \times V \rightarrow [0, \infty]$ induced by $\|\cdot\|$ (Definition C.2.2).

Definition C.2.5 (Extended Norm). Let V be a vector space over a field F equipped with an absolute value (Definition B.1.1)

$$|\cdot| : F \rightarrow [0, \infty).$$

An *extended norm on V* is a function

$$\|\cdot\| : V \rightarrow [0, \infty]$$

satisfying for all $x, y \in V$ and all scalars $\alpha \in F$:

1. **Positive definiteness:** $\|x\| = 0$ if and only if $x = 0$.
2. **Homogeneity:** $\|\alpha x\| = |\alpha| \cdot \|x\|$.
3. **Triangle inequality:** $\|x + y\| \leq \|x\| + \|y\|$,

where arithmetic is extended to allow sums involving ∞ with the convention that $a + \infty = \infty$ for any $a \in [0, \infty]$. A vector space with an extended norm over a field with an absolute value is called an *extended normed (vector) space*.

If the range of the extended norm is contained in $[0, \infty)$, then the extended norm is a *norm* in the usual sense and V may be called a *normed (vector) space*.

REFERENCES

- [Ayo23] Joseph Ayoub. Counterexamples to F. Morel’s conjecture on $\pi_0^{\mathcal{D}^1}$. *Comptes Rendus. Mathématique*, 361:1087–1090, 2023.
- [BBD82] Alexander A. Beilinson, Joseph Bernstein, and Pierre Deligne. Analyse et topologie sur les espaces singuliers (i). *Astérisque*, 100, 1982.
- [BC19] Tilman Bauer and Magnus Carlson. Tensor products of affine and formal abelian groups. *Documenta Mathematica*, 24:2525–2582, 2019.
- [BGI71] Pierre Berthelot, Alexander Grothendieck, and Luc Illusie. *Théorie des Intersections et Théorème de Riemann-Roch (SGA6)*, volume 225 of *Lecture Notes in Mathematics*. Springer-Verlag, 1971.
- [Bor12] Armand Borel. *Linear algebraic groups*, volume 126 of *Graduate Texts in Mathematics*. Springer New York, 2012.
- [BSS18] Bhargav Bhatt, Christian Schnell, and Peter Scholze. Vanishing theorems for perverse sheaves on abelian varieties, revisited. *Selecta Mathematica*, 24:63–84, 2018.
- [Cho08] Utsav Choudhury. Homotopy theory of schemes and a^1 -fundamental groups. Master’s thesis, Università degli Studi di Padova, 2008.

- [DA73] Pierre Deligne and Michael Artin. *Théorie des Topos et Cohomologies Étale des Schémas. Séminaire de Géométrie Algébrique due Bois-Marie 1963-1964 (SGA 4)*. Lecture Notes in Mathematics. Springer Berlin, 1973.
- [DBG⁺77] Pierre Deligne, Jean-François Boutot, Alexander Grothendieck, Luc Illusie, and Jean-Louis Verdier. *Étale Cohomology. Séminaire de Géométrie Algébrique due Bois-Marie 1963-1964 (SGA 4 1/2)*. Lecture Notes in Mathematics. Springer-Verlag, 1977.
- [Del80] Pierre Deligne. La conjecture de Weil : II. *Publications Mathématiques de l’IHÉS*, 52:137–252, 1980.
- [Eke07] Torsten Ekedahl. *On The Adic Formalism*, pages 197–218. Birkhäuser Boston, Boston, MA, 2007.
- [FFK24] Arthur Forey, Javier Fresán, and Emmanuel Kowalski. Arithmetic fourier transforms over finite fields: generic vanishing, convolution, and equidistribution, 2024.
- [Fu15] Lei Fu. *Etale Cohomology Theory*, volume 14 of *Nankai Tracts in Mathematics*. World Scientific, 2015.
- [GL96] Ofer Gabber and François Loeser. Faisceaux pervers ℓ -adiques sur un tore. *Duke Math J.*, 83(3):501–606, 1996.
- [GR04] Alexander Grothendieck and Michèle Raynaud. Revêtements étales et groupe fondamental (SGA 1). eprint arXiv math/0206203, 2004. Updated edition of the book of the same title published by Springer-Verlag in 1971 as volume 224 of the series Lecture Notes in Mathematics.
- [Gro77] Alexander Grothendieck. *Cohomologie ℓ -adique et fonctions L Séminaire de Géométrie Algébrique due Bois-Marie 1965-1966 (SGA 5)*, volume 589 of *Springer Lecture Notes*. Springer-Verlag, 1977. Avec la collaboration de I. Bucur, C. Houzel, L. Illusie, J.-P. Jouanolou, et J.-P. Serre.
- [GV72] Alexander Grothendieck and Jean-Louis Verdier. *Théorie des Topos et Cohomologie Étale des Schémas. Séminaire de Géométrie Algébrique due Bois-Marie 1963-1964 (SGA 4)*. Lecture Notes in mathematics. Springer-Verlag Berlin Heidelberg, 1 edition, 1972.
- [HM73] Dale Husemoller and John Milnor. *Symmetric Bilinear Forms*, volume 73 of *Ergebnisse der Mathematik und ihrer Grenzgebiete. 2. Folge*. Springer Berlin Heidelberg, 1973.
- [Hub97] Annette Huber. Mixed perverse sheaves for schemes over number fields. *Compositio Mathematica*, 108:107–121, 1997.
- [Kat90] Nicholas M. Katz. *Exponential sums and differential equations*, volume 124 of *Annals of Mathematics Studies*. Princeton University Press, 1990.
- [Kat96] Nicholas M. Katz. *Rigid Local Systems*, volume 139 of *annals of Mathematics Studies*. Princeton University Press, 1996.
- [Kat98] Nicholas M. Katz. *Gauss Sums, Kloosterman Sums, and Monodromy Groups*, volume 116 of *Annals of Mathematics Studies*. Princeton University Press, 1998.
- [Kat12] Nicholas M. Katz. *Convolution and Equidistribution Sato-Tate Theorems for Finite-Field Mellin Transforms*, volume 180 of *Annals of Mathematics Studies*. Princeton University Press, 2012.
- [KL85] Nicholas M. Katz and Gérard Laumon. Transformation de fourier et majoration de sommes exponentielles. *Publications Mathématiques de l’IHÉS*, 62:145–202, 1985.
- [Krä14] Thomas Krämer. Perverse sheaves on semiabelian varieties. *Rendiconti del Seminario Matematico della Università di Padova*, 132:83–102, 2014.
- [KW13] Reinhardt Kiehl and Rainer Weissauer. *Weil Conjectures, Perverse Sheaves and ℓ -adic Fourier Transform*, volume 42 of *Ergebnisse der Mathematik und ihrer Grenzgebiete. 3. Folge / A Series of Modern Surveys in Mathematics*. Springer Berlin, Heidelberg, 2013.
- [KW15] Thomas Krämer and Rainer Weissauer. Vanishing theorems for constructible sheaves on abelian varieties. *J. Algebraic Geometry*, 24:531–568, 2015.
- [May99] Jon Peter May. *A Concise Course in Algebraic Topology*. Chicago Lectures in Mathematics. University of Chicago Press, 1999.
- [Mor06] Fabien Morel. A_1 -algebraic topology. In *International Congress of Mathematicians*, volume 2, pages 1035–1059, 2006.
- [Mor12] Fabien Morel. *A_1 -Algebraic Topology over a field*. Lecture Notes in Mathematics. Springer Berlin, Heidelberg, 2012.
- [MV99] Fabien Morel and Vladimir Voevodsky. A_1 -homotopy theory of schemes. *Publications Mathématiques de l’IHÉS*, 90:45–143, 1999.

- [nLa25a] nLab authors. geometric morphism. <https://ncatlab.org/nlab/show/geometric+morphism>, July 2025. Revision 61.
- [nLa25b] nLab authors. homotopy group of a spectrum. <https://ncatlab.org/nlab/show/homotopy+group+of+a+spectrum>, June 2025. Revision 7.
- [nLa25c] nLab authors. Introduction to Stable homotopy theory – 1-1. <https://ncatlab.org/nlab/show/Introduction+to+Stable+homotopy+theory+--+1-1>, June 2025. Revision 43.
- [nLa25d] nLab authors. model structure on topological sequential spectra. <https://ncatlab.org/nlab/show/model+structure+on+topological+sequential+spectra>, June 2025. Revision 61.
- [nLa25e] nLab authors. point of a topos. <https://ncatlab.org/nlab/show/point+of+a+topos>, July 2025. Revision 53.
- [nLa25f] nLab authors. stable homotopy category. <https://ncatlab.org/nlab/show/stable+homotopy+category>, June 2025. Revision 31.
- [Ros02] Michael Rosen. *Number Theory in Function Fields*, volume 210 of *Graduate Texts in Mathematics*. Springer-Verlag New York, 2002.
- [Rud87] Walter Rudin. *Real and Complex Analysis*. Mathematics Series. McGraw-Hill Book Company, 3 edition, 1987.
- [Sta25] The Stacks project authors. The stacks project. <https://stacks.math.columbia.edu>, 2025.
- [Voe98] Vladimir Voevodsky. A1-homotopy theory. In *Proceedings of the international congress of mathematicians*, volume 1, pages 579–604. Berlin, 1998.
- [Wei94] Charles A. Weibel. *An Introduction to Homological Algebra*, volume 38 of *Cambridge Studies in Advanced Mathematics*. Cambridge University Press, 1994. First paperback edition 1995 Reprinted 1997.
- [Wik25] Wikipedia contributors. Frobenius endomorphisms#frobenius for schemes — Wikipedia, the free encyclopedia, 2025. [Online; accessed 08-July-2025].